定义$SE_2(3)$矩阵李群变量：

$$\chi = \begin{bmatrix} C_b^e & v_{eb}^e & r_{eb}^e \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix} \in SE_2(3) \tag{1}$$

式中，C_b^e为载体坐标系 b 在 ECEF 坐标系下的姿态 DCM；v_{eb}^e为载体坐标系 b 在 ECEF 坐标系下的速度，r_{eb}^e为载体坐标系 b 在 ECEF 坐标系下的坐标；

χ满足如下关系式：

$$\chi = \exp_G(\Lambda(\zeta)) = \sum_{n=0}^{\infty} \frac{1}{n!}(\Lambda(\zeta))^n = \exp_G \begin{pmatrix} \phi^\times & \theta & \xi \\ 0_{1\times3} & 0 & 0 \\ 0_{1\times3} & 0 & 0 \end{pmatrix} = \begin{bmatrix} \exp_G(\phi^\times) & J\theta & J\xi \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix} \tag{2}$$

式中，$\Lambda(\zeta) \in \mathfrak{se}_2(3)$表示$SE_2(3)$对应的李代数，$\zeta = \begin{bmatrix} \phi \\ \theta \\ \xi \end{bmatrix} \in \mathbb{R}^9$，$\phi, \theta, \xi \in \mathbb{R}^3$，$\exp_G$表示矩阵指数运算；

$$\exp_G(\phi^\times) = \sum_{n=0}^{\infty} \frac{1}{n!}(\phi^\times)^n = I_{3\times3} + \frac{\sin(\varphi)}{\varphi}\phi^\times + \frac{1-\cos(\varphi)}{\varphi^2}(\phi^\times)^2, \varphi = \|\phi\| \tag{3}$$

$$J = J_l(\phi) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!}(\phi^\times)^n = I_{3\times3} + \frac{1-\cos(\varphi)}{\varphi^2}\phi^\times + \frac{\varphi-\sin(\varphi)}{\varphi^3}(\phi^\times)^2 \tag{4}$$

$$\chi = \sum_{n=0}^{\infty} \frac{1}{n!}(\Lambda(\zeta))^n = I_{5\times5} + \Lambda(\zeta) + \frac{1-\cos(\varphi)}{\varphi^2}(\Lambda(\zeta))^2 + \frac{\varphi-\sin(\varphi)}{\varphi^3}(\Lambda(\zeta))^3 \tag{5}$$

上式中，$J_l(\phi)$为左雅可比矩阵；

χ左乘和右乘运算为

$$\chi_l = \chi\exp_G(\Lambda(\zeta_l)) = \chi\exp_G(\Lambda(\zeta_l))\chi^{-1}\chi = \exp_G\left(\Lambda\left(\mathrm{Ad}_\chi(\zeta_l)\right)\right)\chi \tag{6}$$

$$\chi_r = \exp_G(\Lambda(\zeta_r))\chi \tag{7}$$

式中，Ad_χ为群$SE_2(3)$在χ处的伴随矩阵，即

$$\mathrm{Ad}_\chi(\zeta_l) = \Lambda^{-1}(\chi\Lambda(\zeta_l)\chi^{-1}) \tag{8}$$

$$\Lambda^{-1}(\Theta) = \vartheta \in \mathbb{R}^9, \Theta \in \mathfrak{se}_2(3), \Lambda(\vartheta) = \Theta \tag{9}$$

由此可得

$$\mathrm{Ad}_\chi(\zeta_l) = \begin{bmatrix} C_b^e & 0 & 0 \\ (v_{eb}^e)^\times C_b^e & C_b^e & 0 \\ (r_{eb}^e)^\times C_b^e & 0 & C_b^e \end{bmatrix} \cdot \zeta_l \tag{10}$$

$$\chi^{-1} = \begin{bmatrix} C_b^e & v_{eb}^e & r_{eb}^e \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} C_e^b & -C_e^b v_{eb}^e & -C_e^b r_{eb}^e \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix} = \begin{bmatrix} C_e^b & -v_{eb}^b & -r_{eb}^b \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix} \in SE_2(3) \tag{11}$$

$$\dot{\chi}=\begin{bmatrix}\dot{C}_b^e & \dot{v}_{eb}^e & \dot{r}_{eb}^e\\ 0_{1\times3} & 1 & 0\\ 0_{1\times3} & 0 & 1\end{bmatrix}=\begin{bmatrix}C_b^e\omega_{ib}^{b\times}-\omega_{ie}^{e\times}C_b^e & C_b^e f_{ib}^b+g^e-2\omega_{ie}^{e\times}v_{eb}^e-\omega_{ie}^{e\times}\omega_{ie}^{e\times}r_{eb}^e & v_{eb}^e\\ 0_{1\times3} & 1 & 0\\ 0_{1\times3} & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}C_b^e & v_{eb}^e & r_{eb}^e\\ 0_{1\times3} & 1 & 0\\ 0_{1\times3} & 0 & 1\end{bmatrix}\underbrace{\begin{bmatrix}\omega_{ib}^{b\times} & f_{ib}^b & 0\\ 0_{1\times3} & 0 & 0\\ 0_{1\times3} & 0 & 0\end{bmatrix}}_{W_1}$$

$$+\underbrace{\begin{bmatrix}-\omega_{ie}^{e\times} & -\omega_{ie}^{e\times}v_{eb}^e+g^e-\omega_{ie}^{e\times}\omega_{ie}^{e\times}r_{eb}^e & v_{eb}^e+\omega_{ie}^{e\times}r_{eb}^e\\ 0_{1\times3} & 0 & 0\\ 0_{1\times3} & 0 & 0\end{bmatrix}}_{W_2}\begin{bmatrix}C_b^e & v_{eb}^e & r_{eb}^e\\ 0_{1\times3} & 1 & 0\\ 0_{1\times3} & 0 & 1\end{bmatrix}$$

$$=\chi W_1+W_2\chi \tag{12}$$

定义误差项：

$$\varepsilon=\chi^{-1}\tilde{\chi}=\begin{bmatrix}C_e^b\tilde{C}_b^e & C_e^b(\tilde{v}_{eb}^e-v_{eb}^e) & C_e^b(\tilde{r}_{eb}^e-r_{eb}^e)\\ 0_{1\times3} & 1 & 0\\ 0_{1\times3} & 0 & 1\end{bmatrix}=\begin{bmatrix}\exp_G(\varepsilon^{a\times}) & \varepsilon^v & \varepsilon^r\\ 0_{1\times3} & 1 & 0\\ 0_{1\times3} & 0 & 1\end{bmatrix} \tag{13}$$

$$\exp_G(\varepsilon^{a\times})=C_e^b\tilde{C}_b^e \tag{14}$$

$$\varepsilon^v=C_e^b\delta v_{eb}^e,\delta v_{eb}^e=\tilde{v}_{eb}^e-v_{eb}^e \tag{15}$$

$$\varepsilon^r=C_e^b\delta r_{eb}^e,\delta r_{eb}^e=\tilde{r}_{eb}^e-r_{eb}^e \tag{16}$$

对参数ε^a、ε^v和ε^r求关于时间的导数：

$$\frac{d}{dt}\exp_G(\varepsilon^{a\times})=\frac{d}{dt}\left(C_e^b\tilde{C}_b^e\right)=\dot{C}_e^b\tilde{C}_b^e+C_e^b\dot{\tilde{C}}_b^e=\left(C_e^b\omega_{ie}^{e\times}-\omega_{ib}^{b\times}C_e^b\right)\tilde{C}_b^e+C_e^b\left(\tilde{C}_b^e\tilde{\omega}_{ib}^{b\times}-\omega_{ie}^{e\times}\tilde{C}_b^e\right)$$
$$=C_e^b\omega_{ie}^{e\times}\tilde{C}_b^e-\omega_{ib}^{b\times}C_e^b\tilde{C}_b^e+C_e^b\tilde{C}_b^e\tilde{\omega}_{ib}^{b\times}-C_e^b\omega_{ie}^{e\times}\tilde{C}_b^e=-\omega_{ib}^{b\times}C_e^b\tilde{C}_b^e+C_e^b\tilde{C}_b^e\tilde{\omega}_{ib}^{b\times}$$
$$=-\omega_{ib}^{b\times}\exp_G(\varepsilon^{a\times})+\exp_G(\varepsilon^{a\times})\tilde{\omega}_{ib}^{b\times} \tag{17}$$

考虑到$\tilde{\omega}_{ib}^b=\omega_{ib}^b+\delta\omega_{ib}^b$，则有

$$\frac{d}{dt}\exp_G(\varepsilon^{a\times})=-\left(\tilde{\omega}_{ib}^b-\delta\omega_{ib}^b\right)^{\times}\exp_G(\varepsilon^{a\times})+\exp_G(\varepsilon^{a\times})\tilde{\omega}_{ib}^{b\times}$$
$$=-\tilde{\omega}_{ib}^{b\times}\exp_G(\varepsilon^{a\times})+\left(\delta\omega_{ib}^b\right)^{\times}\exp_G(\varepsilon^{a\times})+\exp_G(\varepsilon^{a\times})\tilde{\omega}_{ib}^{b\times} \tag{18}$$

当$\|\varepsilon^a\|$比较小时，则有

$$\exp_G(\varepsilon^{a\times})\approx I_{3\times3}+\varepsilon^{a\times} \tag{19}$$

$$\frac{d}{dt}\exp_G(\varepsilon^{a\times})\approx\frac{d}{dt}(\varepsilon^{a\times})=-\tilde{\omega}_{ib}^{b\times}(I_{3\times3}+\varepsilon^{a\times})+\left(\delta\omega_{ib}^b\right)^{\times}(I_{3\times3}+\varepsilon^{a\times})+(I_{3\times3}+\varepsilon^{a\times})\tilde{\omega}_{ib}^{b\times}$$
$$=-\tilde{\omega}_{ib}^{b\times}-\tilde{\omega}_{ib}^{b\times}\varepsilon^{a\times}+\left(\delta\omega_{ib}^b\right)^{\times}+\left(\delta\omega_{ib}^b\right)^{\times}\varepsilon^{a\times}+\tilde{\omega}_{ib}^{b\times}+\varepsilon^{a\times}\tilde{\omega}_{ib}^{b\times}$$
$$=\left(\delta\omega_{ib}^b\right)^{\times}+\varepsilon^{a\times}\tilde{\omega}_{ib}^{b\times}-\tilde{\omega}_{ib}^{b\times}\varepsilon^{a\times}+\left(\delta\omega_{ib}^b\right)^{\times}\varepsilon^{a\times}\approx\left(\delta\omega_{ib}^b\right)^{\times}+\left(\varepsilon^{a\times}\tilde{\omega}_{ib}^b\right)^{\times} \tag{20}$$

由此可得

$$\dot{\varepsilon}^a=-\left(\tilde{\omega}_{ib}^b\right)^{\times}\varepsilon^a+\delta\omega_{ib}^b=-\left(\tilde{\omega}_{ib}^b\right)^{\times}\varepsilon^a+b_w+\eta_g \tag{21}$$

$$\frac{d}{dt}\varepsilon^v = \frac{d}{dt}(C_e^b \delta v_{eb}^e) = \dot{C}_e^b \delta v_{eb}^e + C_e^b \delta \dot{v}_{eb}^e$$

$$= \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right)\delta v_{eb}^e$$

$$+ C_e^b\left(\tilde{C}_b^e \tilde{f}_{ib}^b + g^e - 2\omega_{ie}^{e\times}\tilde{v}_{eb}^e - \omega_{ie}^{e\times}\omega_{ie}^{e\times}\tilde{r}_{eb}^e\right.$$

$$\left. - \left(C_b^e f_{ib}^b + g^e - 2\omega_{ie}^{e\times} v_{eb}^e - \omega_{ie}^{e\times}\omega_{ie}^{e\times} r_{eb}^e\right)\right)$$

$$= \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right)\delta v_{eb}^e$$

$$+ C_e^b\left(\tilde{C}_b^e \tilde{f}_{ib}^b - C_b^e f_{ib}^b - 2\omega_{ie}^{e\times}\delta v_{eb}^e - \omega_{ie}^{e\times}\omega_{ie}^{e\times}\delta r_{eb}^e\right)$$

$$= \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right)\delta v_{eb}^e + C_e^b \tilde{C}_b^e \tilde{f}_{ib}^b - f_{ib}^b - 2C_e^b \omega_{ie}^{e\times}\delta v_{eb}^e$$

$$- C_e^b \omega_{ie}^{e\times}\omega_{ie}^{e\times}\delta r_{eb}^e$$

$$= \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right) C_b^e \varepsilon^v + C_e^b \tilde{C}_b^e \tilde{f}_{ib}^b - f_{ib}^b - 2C_e^b \omega_{ie}^{e\times} C_b^e \varepsilon^v$$

$$- C_e^b \omega_{ie}^{e\times}\omega_{ie}^{e\times} C_b^e \varepsilon^r$$

$$= C_e^b \omega_{ie}^{e\times} C_b^e \varepsilon^v - \omega_{ib}^{b\times}\varepsilon^v + C_e^b \tilde{C}_b^e \tilde{f}_{ib}^b - f_{ib}^b - 2C_e^b \omega_{ie}^{e\times} C_b^e \varepsilon^v - C_e^b \omega_{ie}^{e\times}\omega_{ie}^{e\times} C_b^e \varepsilon^r$$

$$\approx \left(\tilde{C}_e^b \omega_{ie}^e\right)^\times \varepsilon^v - \tilde{\omega}_{ib}^{b\times}\varepsilon^v + (I_{3\times 3} + \varepsilon^{a\times})\tilde{f}_{ib}^b - \tilde{f}_{ib}^b + \delta f_{ib}^b - 2\tilde{C}_e^b \omega_{ie}^{e\times} C_b^e \varepsilon^v$$

$$- \tilde{C}_e^b \omega_{ie}^{e\times}\omega_{ie}^{e\times}\tilde{C}_b^e \varepsilon^r$$

$$= -\left(\tilde{C}_e^b \omega_{ie}^e\right)^\times \varepsilon^v - \tilde{\omega}_{ib}^{b\times}\varepsilon^v - \left(\tilde{f}_{ib}^b\right)^\times \varepsilon^a + \delta f_{ib}^b - \tilde{C}_e^b \omega_{ie}^{e\times}\omega_{ie}^{e\times}\tilde{C}_b^e \varepsilon^r \tag{22}$$

由此可得

$$\dot{\varepsilon}^v = -\left(\tilde{C}_e^b \omega_{ie}^e + \tilde{\omega}_{ib}^b\right)^\times \varepsilon^v - \tilde{f}_{ib}^{b\times}\varepsilon^a - \tilde{C}_e^b \omega_{ie}^{e\times}\omega_{ie}^{e\times}\tilde{C}_b^e \varepsilon^r + b_a + \eta_a \tag{23}$$

$$\frac{d}{dt}\varepsilon^r = \frac{d}{dt}(C_e^b \delta r_{eb}^e) = \dot{C}_e^b \delta r_{eb}^e + C_e^b \delta \dot{r}_{eb}^e = \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right)\delta r_{eb}^e + C_e^b \delta v_{eb}^e$$

$$= \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right) C_b^e \varepsilon^r + C_e^b C_b^e \varepsilon^v = \left(C_e^b \omega_{ie}^{e\times} - \omega_{ib}^{b\times} C_e^b\right) C_b^e \varepsilon^r + \varepsilon^v$$

$$= C_e^b \omega_{ie}^{e\times} C_b^e \varepsilon^r - \omega_{ib}^{b\times}\varepsilon^r + \varepsilon^v \approx \tilde{C}_e^b \omega_{ie}^{e\times}\tilde{C}_b^e \varepsilon^r - \left(\tilde{\omega}_{ib}^b\right)^\times \varepsilon^r + \left(\delta\omega_{ib}^b\right)^\times \varepsilon^r + \varepsilon^v$$

$$\approx \tilde{C}_e^b \omega_{ie}^{e\times}\tilde{C}_b^e \varepsilon^r - \left(\tilde{\omega}_{ib}^b\right)^\times \varepsilon^r + \varepsilon^v = -\left(\tilde{\omega}_{ib}^b - \tilde{C}_e^b \omega_{ie}^e\right)^\times \varepsilon^r + \varepsilon^v \tag{24}$$

IMU 加速度和陀螺偏差建模为一阶马尔科夫模型，即

$$\dot{b}_a = -\frac{1}{\tau_a} b_a + \mu_a \tag{25}$$

$$\dot{b}_g = -\frac{1}{\tau_g} b_g + \mu_g \tag{26}$$

定义误差状态参数：

$$\epsilon = \begin{bmatrix} \varepsilon^a \\ \varepsilon^v \\ \varepsilon^r \\ b_g \\ b_a \end{bmatrix} \in \mathbb{R}^{15} \tag{27}$$

$$\dot{\epsilon} = \begin{bmatrix} -\left(\tilde{\omega}_{ib}^b\right)^\times & 0 & 0 & \mathrm{I}_{3\times3} & 0 \\ -\left(\tilde{f}_{ib}^b\right)^\times & -\left(\tilde{\mathrm{C}}_e^b\omega_{ie}^e + \tilde{\omega}_{ib}^b\right)^\times & -\tilde{\mathrm{C}}_e^b\omega_{ie}^{e\,\times}\omega_{ie}^{e\,\times}\tilde{\mathrm{C}}_b^e & 0 & \mathrm{I}_{3\times3} \\ 0 & \mathrm{I}_{3\times3} & -\left(\tilde{\omega}_{ib}^b - \tilde{\mathrm{C}}_e^b\omega_{ie}^e\right)^\times & 0 & 0 \\ 0 & 0 & 0 & -\dfrac{\mathrm{I}_{3\times3}}{\tau_g} & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{\mathrm{I}_{3\times3}}{\tau_a} \end{bmatrix} \begin{bmatrix} \varepsilon^a \\ \varepsilon^v \\ \varepsilon^r \\ b_g \\ b_a \end{bmatrix} + \begin{bmatrix} \mathrm{I}_{3\times3} & 0 & 0 & 0 \\ 0 & \mathrm{I}_{3\times3} & 0 & 0 \\ 0 & 0 & \mathrm{I}_{3\times3} & 0 \\ 0 & 0 & 0 & \mathrm{I}_{3\times3} \end{bmatrix} \begin{bmatrix} \eta_g \\ \eta_a \\ \mu_g \\ \mu_a \end{bmatrix} \tag{28}$$

假设有位置观测值为$\hat{r}_{eb}^e$，则有

$$y = \begin{bmatrix} \hat{r}_{ea}^e \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathrm{C}_b^e & v_{eb}^e & r_{eb}^e \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix} \begin{bmatrix} l_{ba}^b \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} n \\ 0 \\ 0 \end{bmatrix} \tag{29}$$

$$\mathrm{y} = \chi\mathrm{b} + \mathrm{V} \tag{30}$$

式中，l_{ba}^b为杆臂，n 为观测噪声；

$$\mathrm{z} = \tilde{\chi}^{-1}\mathrm{y} - \tilde{\chi}^{-1}(\chi\mathrm{b} + \mathrm{V}) = \begin{bmatrix} -\tilde{\mathrm{C}}_b^e\left(l_{ba}^b\right)^\times & 0 & \tilde{\mathrm{C}}_b^e & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \varepsilon^a \\ \varepsilon^v \\ \varepsilon^r \\ b_g \\ b_a \end{bmatrix} + \begin{bmatrix} \tilde{\mathrm{C}}_e^b n \\ 0 \\ 0 \end{bmatrix} \tag{31}$$